LECTURE 5:
INFERENCE FOR NORMAL MEANS AND VARIANCES

Key concepts:

- Student t distribution
- Marginal posterior distribution
- Predictive distribution

1. INTRODUCTION

These notes combine the results from lectures 3 and 4 to perform inferences on both the mean and variance of a normal distribution. There are no new ideas in this lecture, but we start to see the complexity that arises when we attempt real applications. Almost all of the details are provided for completeness, but we will focus on how these results relate to one another and skip most of the proofs.

2. STUDENT t DISTRIBUTION

In elementary statistics, you encountered Student's t distribution in connection with the t-statistic for testing hypotheses about sample means and in the formation of confidence intervals. We will briefly summarize these results.

Definition 1. *If t is a continuous random variable with pdf*

$$p(t) = \frac{\Gamma\left(\dfrac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\dfrac{\nu}{2}\right)}\left(1+\frac{t^2}{\nu}\right)^{-(\nu+1)/2},$$

then t is said to have Student's t distribution *with ν degrees of freedom and we write $t \sim t(\nu)$.*

In addition, we can define a location-scale family by letting $y = \mu + \sigma t$ where $t \sim t(\nu)$. Then

$$p_y(y) = p_t\left(\frac{y-\mu}{\sigma}\right)\left|\frac{1}{\sigma}\right| = \frac{\Gamma\left(\dfrac{\nu+1}{2}\right)}{\sigma\sqrt{\nu\pi}\,\Gamma\left(\dfrac{\nu}{2}\right)}\left[1+\frac{1}{\nu}\left(\frac{y-\mu}{\sigma}\right)^2\right]^{-(\nu+1)/2}$$

In this case, we write $y \sim t_\nu(\mu, \sigma^2)$ (so $t(\nu)$ is shorthand for $t_\nu(0,1)$).

The t distribution plays an important part in finite sample frequentist inference with normal data. The following result is fundamental, though the proof is just a routine calculation.

Proposition 1. *If $z \sim$ Normal$(0,1)$ and $y \sim \chi^2(\nu)$ are independent random variables, then*

$$t = \frac{z}{\sqrt{y/\nu}} \sim t(\nu).$$

Proof. Consider the one-to-one transformation $(z, y) \mapsto (t, y)$ with (inverse) Jacobian $\sqrt{y/\nu}$, so

$$\begin{aligned} p_{ty}(t, y) &= p_{zy}\Big(t\sqrt{y/\nu}, y\Big)\ \sqrt{y/\nu} \\ &= \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{1}{2}t^2 y/\nu\right\} \frac{(1/2)^{-\nu/2}}{\Gamma(\nu/2)} y^{\nu/2} e^{-y/2}\sqrt{y/\nu} \\ &= \frac{(1/2)^{-(\nu+1)/2}}{\sqrt{\nu\pi}\,\Gamma(\nu/2)} y^{(\nu+1)/2-1} \exp\left\{-\frac{1}{2}\left(1+\frac{t^2}{\nu}\right) y\right\}. \end{aligned}$$

Let $g(t) = (1 + t^2/\nu)/2$. The marginal distribution of t is

$$\begin{aligned} p_t(t) = \int_0^\infty p_{ty}(t, y)\, dy &= \frac{(1/2)^{-(\nu+1)/2}}{\sqrt{\nu\pi}\,\Gamma(\nu/2)} \int_0^\infty y^{(\nu+1)/2-1} e^{-g(t)y} dy \\ &= \frac{(1/2)^{-(\nu+1)/2}}{\sqrt{\nu\pi}\,\Gamma(\nu/2)} \frac{\Gamma((\nu+1)/2)}{g(t)^{(\nu+1)/2}} = \frac{\Gamma\left(\dfrac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\dfrac{\nu}{2}\right)} \left(1+\frac{t^2}{\nu}\right)^{-(\nu+1)/2}. \end{aligned}$$

□

In your regression course, you will have seen the sampling distribution of the sample mean and the sample sum of squares: $\bar{y} \sim \text{Normal}(\mu, \sigma^2/n)$ and $(n-1)s^2/\sigma^2 \sim \chi^2(n-1)$ are independent random variables, so that

$$t = \frac{\dfrac{\bar{y}-\mu}{\sigma/\sqrt{n}}}{\sqrt{\dfrac{(n-1)s^2/\sigma^2}{n-1}}} = \frac{\bar{y}-\mu}{s/\sqrt{n}} \sim t(n-1).$$

We will see shortly that Bayesian inference with a particular prior distribution yields the identical result, except with μ being random (*i.e.*, unknown) and $\bar{y}$ as fixed (*i.e.*, known)!

3. Noninformative Prior

We start by considering inference for the mean and variance using a "noninformative" prior. The range of the parameters $(\mu, \log\sigma^2)$ is the entire plane, so we might try an improper uniform prior on the plane,

$$p_{\mu,\log\sigma^2}(\mu, \log\sigma^2) \propto 1,$$

as a noninformative prior. This is equivalent to

$$p_{\mu,\sigma^2}(\mu, \sigma^2) = p_{\mu,\log\sigma^2}(\mu, \log\sigma^2) \left|\frac{\partial \log\sigma^2}{\partial\sigma^2}\right| \propto \frac{1}{\sigma^2}.$$

Proposition 2. *Suppose $y_1, \ldots, y_n$ are i.i.d. Normal(μ, σ^2), where the prior is $p(\mu, \sigma^2) \propto 1/\sigma^2$.*

(a) $\sigma^2|y \sim$ *Scaled Inverse Chi-squared*$(n-1, s^2)$

(b) $\mu|\sigma^2, y \sim$ *Normal*$(\bar{y}, \sigma^2/n)$

(c) $\mu|y \sim t_{n-1}(\bar{y}, s^2/n)$

Comment. The conditional posterior distribution of μ given σ^2 is exactly what was obtained in Lecture 3 treating σ^2 as known. The posterior distribution of σ^2 is identical to that discussed in Lecture 4 treating μ as known. The most notable result, however, is that the marginal posterior distribution of μ is a Student t distribution centered on the sample mean $\bar{y}$ and with scale equal to the usual estimate of the standard error of $\bar{y}$,

$$\widehat{\text{s.e.}}(\bar{y}) = \frac{s}{\sqrt{n}},$$

where $s^2 = \sum_{i=1}^n (y_i - \bar{y})^2/(n-1)$ is the usual unbiased variance estimator. The primary difference here is that this is *not* a sampling distribution. The sample mean $\bar{y}$ and the sample standard deviation s are functions of the sample data $y = (y_1, \dots, y_n)$ and are being conditioned upon. The random variables are the parameters μ and σ^2.

Proof. The joint posterior is

$$\begin{aligned}
p(\mu, \sigma^2|y) &\propto p(\mu, \sigma^2) p(y_1, \dots, y_n|\mu, \sigma^2) \\
&\propto \frac{1}{\sigma^2} (\sigma^2)^{-n/2} \exp\left\{ -\frac{(n-1)s^2 + n(\bar{y} - \mu)^2}{2\sigma^2} \right\} \\
&\propto (\sigma^2)^{-(n+1)/2} \exp\left\{ -\frac{(n-1)s^2}{2\sigma^2} \right\} (\sigma^2/n)^{-1/2} \exp\left\{ -\frac{(\mu - \bar{y})^2}{2\sigma^2/n} \right\}
\end{aligned}$$

To find the marginal posterior distribution for σ^2, integrate out μ:

$$\begin{aligned}
p(\sigma^2|y) &= \int_{-\infty}^{\infty} p(\mu, \sigma^2|y)\, d\mu \\
&\propto (\sigma^2)^{-(n+1)/2} \exp\left\{ -\frac{(n-1)s^2}{2\sigma^2} \right\} (\sigma^2/n)^{-1/2} \int_{-\infty}^{\infty} \exp\left\{ -\frac{(\mu - \bar{y})^2}{2\sigma^2/n} \right\} d\mu \\
&\propto (\sigma^2)^{-(n+1)/2} \exp\left\{ -\frac{(n-1)s^2}{2\sigma^2} \right\} \sqrt{2\pi},
\end{aligned}$$

which is proportional to a Scaled Inverse Chi-squared$(n-1, s^2)$ pdf. This proves part (a).

For part (b),

$$p(\mu|\sigma^2, y) = \frac{p(\mu, \sigma^2|y)}{p(\sigma^2|y)} \propto (\sigma^2/n)^{-1/2} \exp\left\{ -\frac{(\mu - \bar{y})^2}{2\sigma^2/n} \right\},$$

which is a Normal$(\bar{y}, s^2/n)$ pdf.

Finally, for part (c), integrate out σ^2 from the joint posterior:

$$\begin{aligned}
p(\mu|y) &= \int_0^{\infty} p(\mu, \sigma^2)\, d\sigma^2 \\
&\propto \int_0^{\infty} (\sigma^2)^{-n/2-1} \exp\left\{ -\frac{(n-1)s^2 + n(\bar{y} - \mu)^2}{2\sigma^2} \right\} d\sigma^2 \\
&= \int_0^{\infty} u^{-\alpha-1} e^{-\beta/u}\, du = \frac{\Gamma(\alpha)}{\beta^\alpha},
\end{aligned}$$

where $\alpha = n/2$ and $\beta = [(n-1)s^2 + n(\bar{y}-\mu)^2]/2$. The final expression in the last line is the reciprocal of the normalizing constant from an Inverse Gamma(α, β) distribution (see Lecture 4, Section 4). Substituting for α and β,

$$p(\mu|y) \propto \beta^{-\alpha} \propto \left[(n-1)s^2 + n(\bar{y}-\mu)^2\right]^{-n/2} \propto \left[1 + \frac{(\bar{y}-\mu)^2}{(n-1)s^2/n}\right]^{-n/2},$$

which is the kernel of a $t_{n-1}(\bar{y}, s^2/n)$ pdf. □

4. Conjugate Prior for Normal Mean and Variance

We repeat the analysis of the previous section using an informative conjugate prior, but with some unpleasant algebra. The conjugate prior incorporates dependence between the mean and variance by specifying the prior distribution for μ conditionally upon σ^2,

$$\mu|\sigma^2 \sim \text{Normal}\left(\mu_0, \frac{\sigma^2}{\kappa_0}\right).$$

The variance σ^2/κ_0 of the prior for μ is specified as being being proportional to the variance σ^2 of the data y, with κ_0 being the number of "prior equivalent observations."

Proposition 3. *Suppose $y_1, \ldots, y_n$ are i.i.d. Normal(μ, σ^2) where the prior is*

$$\begin{aligned}
\sigma^2 &\sim \textit{Scaled Inverse Chi-squared}(\nu_0, \sigma_0^2) \\
\mu|\sigma^2 &\sim \textit{Normal}(\mu, \sigma^2/\kappa_0)
\end{aligned}$$

Then the posterior distribution of (μ, σ^2) is

$$\begin{aligned}
\sigma^2|y &\sim \textit{Scaled Inverse Chi-squared}(\nu_n, \sigma_n^2) \\
\mu|\sigma^2, y &\sim \textit{Normal}(\mu_n, \sigma^2/\kappa_n)
\end{aligned}$$

where

$$\begin{aligned}
\mu_n &= \frac{\kappa_0}{\kappa_0+n}\mu_0 + \frac{n}{\kappa_0+n}\bar{y} \\
\kappa_n &= \kappa_0 + n \\
\nu_n &= \nu_0 + n \\
\sigma_n^2 &= \frac{\nu_0\sigma_0^2 + (n-1)s^2}{\nu_0+n} + \frac{\kappa_0 n(\bar{y}-\mu_0)^2}{(\nu_0+n)(\kappa_0+n)}
\end{aligned}$$

Proof. The proof is similar to that for Proposition 2. The prior pdfs are

$$\begin{aligned}
p(\sigma^2) &\propto \left(\sigma^2\right)^{-(\nu_0/2+1)} \exp\left\{-\frac{\nu_0\sigma_0^2}{2\sigma^2}\right\} \\
p(\mu|\sigma^2) &\propto \left(\frac{\sigma^2}{\kappa_0}\right)^{-1/2} \exp\left\{-\frac{(\mu-\mu_0)^2}{2\sigma^2/\kappa_0}\right\}
\end{aligned}$$

The joint posterior pdf is

$$
\begin{aligned}
p(\mu, \sigma^2|y) &\propto p(\sigma^2)\, p(\mu|\sigma^2)\, p(y_1, \ldots, y_n|\mu, \sigma^2) \\
&\propto \left(\sigma^2\right)^{-(\nu_0/2+2)} \exp\left\{-\frac{\nu_0\sigma_0^2 + \kappa_0(\mu-\mu_0)^2}{2\sigma^2}\right\} \\
&\quad \times \left(\sigma^2\right)^{-n/2} \exp\left\{\frac{(n-1)s^2 + n(\bar{y}-\mu)^2}{2\sigma^2}\right\} \\
&\propto \left(\sigma^2\right)^{-(\nu_0+n)/2-1} \exp\left\{-\frac{\nu_0\sigma_0^2 + (n-1)s^2 + \kappa_0\mu_0^2 + n\bar{y}^2}{2\sigma^2}\right\} \\
&\quad \times \left(\sigma^2\right)^{-1/2} \exp\left\{-\frac{(\kappa_0+n)\mu^2 - 2(\kappa_0\mu_0 + n\bar{y})\mu}{2\sigma^2}\right\} \\
&= \left(\sigma^2\right)^{-(\nu_n)/2-1} \exp\left\{-\frac{\nu_n\sigma_n^2 - \kappa_0 n(\bar{y}-\mu_0)^2/\kappa_n + \kappa_0\mu_0^2 + n\bar{y}^2 - \kappa_n\mu_n^2}{2\sigma^2}\right\} \\
&\quad \times \left(\sigma^2\right)^{-1/2} \exp\left\{-\frac{\kappa_n\mu^2 - 2\kappa_n\mu_n\mu + \kappa_n\mu_n^2}{2\sigma^2}\right\} \\
&= \left(\sigma^2\right)^{-(\nu_n)/2-1} \exp\left\{-\frac{\nu_n\sigma_n^2}{2\sigma^2}\right\} \left(\sigma^2\right)^{-1/2} \exp\left\{-\frac{(\mu-\mu_n)^2}{2\sigma^2/\kappa_n}\right\}
\end{aligned}
$$

since

$$
\begin{aligned}
-\frac{\kappa_0 n(\bar{y}-\mu_0)^2}{\kappa_n} + \kappa_0\mu_0^2 + n\bar{y}^2 - \kappa_n\mu_n^2 &= \frac{\kappa_0(\kappa_n - n)\mu_0^2 + n(\kappa_n-\kappa_0)\bar{y}^2 + 2\kappa_0 n\bar{y}\mu_0 - \kappa_n^2\mu_n^2}{\kappa_n} \\
&= \frac{\kappa_0^2\mu_0^2 + 2n\bar{y}\kappa_0\mu_0 + n^2\bar{y}^2 - \kappa_n^2\mu_n^2}{\kappa_n} = \frac{(\kappa_0\mu_0 + n\bar{y})^2 - \kappa_n^2\mu_n^2}{\kappa_n} = 0.
\end{aligned}
$$

This shows that the posterior is the product of a marginal Scaled Inverse Chi-square(ν_n, σ_n^2) distribution for σ^2 and a conditional Normal($\mu_n, \sigma^2/\kappa_n$) distribution for μ. □

Comment: Notice that taking $\kappa_0, \nu_0 \to 0$ gives the posterior $\sigma^2 \sim$ Scaled Inverse Chi-square($n, \hat{\sigma}^2$) and $\mu|\sigma \sim$ Normal($\bar{y}, \sigma^2/n$) (where $\hat{\sigma}^2 = (n-1)s^2/n$ is the MLE of σ^2), which is slightly different from the posterior starting with the noninformative improper prior. This is a case where the limit of the conjugate prior does *not* yield the noninformative improper prior.

Corollary 1. *With the conjugate prior in Proposition 3, the marginal posterior distribution of* μ *is* $t_{\nu_n}(\mu_n, \sigma_n^2/n)$.

Proof. Identical to the proof of Proposition 2(c). □

5. Posterior Predictive Distribution

Next, we consider the distribution of an additional observations, say $\tilde{y}$, conditional upon the observed data $y = (y_1, \ldots, y_n)$. We shall assume that $\tilde{y}$ is generated independently by the same process and parameters as the observations. That is,

$$
p(\tilde{y}|\mu, \sigma^2, y) = p(\tilde{y}|\mu, \sigma^2) \propto \left(\sigma^2\right)^{-1/2} \exp\left\{-\frac{(\tilde{y}-\mu)^2}{2\sigma^2}\right\}.
$$

so, conditional upon the parameters μ and σ^2, there is no additional information in the observations y. However, our information about the parameters given the observed data y, is represented by the posterior distribution

$$p(\mu, \sigma^2|y) = p(\sigma^2|y)\, p(\mu|\sigma^2, y)$$

where $\sigma^2|y \sim$ Scaled Inverse Chi-squared(ν_n, σ_n^2) and $\mu|\sigma^2, y \sim$ Normal$(\mu_n, \sigma^2/\kappa_n)$.

We want to find the *predictive distribution* of $\tilde{y}$ given y. We need to integrate out the parameters μ and σ^2. We do this in two steps. First,

$$p(\tilde{y}|\sigma^2, y) = \int_{-\infty}^{\infty} p(\tilde{y}|\mu, \sigma^2, y)\, d\mu = \int_{-\infty}^{\infty} p(\tilde{y}|\mu, \sigma^2)\, p(\mu|\sigma^2, y)\, d\mu.$$

Since $\tilde{y}|\mu, \sigma^2 \sim$ Normal(μ, σ^2) and $\mu|\sigma^2, y \sim$ Normal$(\mu_n, \sigma^2/\kappa_n)$, it's easy to show that

$$\int_{-\infty}^{\infty} p(\tilde{y}|\mu, \sigma^2)\, p(\mu|\sigma^2, y)\, d\mu \propto \left(\frac{\sigma^2}{\kappa_n} + \sigma^2\right)^{-1/2} \exp\left\{-\frac{(\tilde{y}-\mu_n)^2}{2\sigma^2(1+1/\kappa_n)}\right\},$$

i.e., a Normal$(\mu_n, (1+1/\kappa_n)\sigma^2)$ distribution.

In the second step, we integrate out σ^2 using the same technique as Propositions 2 and 3:

$$p(\tilde{y}|y) = \int_0^{\infty} p(\tilde{y}|\sigma^2, y)\, p(\sigma^2|y)\, d\sigma^2$$

to obtain a $t_{\nu_n}(\mu_n, (1+\kappa_n)\sigma_n^2)$ distribution. It's a good exercise to fill in the missing details.